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Assignment 4

METCS544A2_S2024

Please type or handwrite your answers clearly and concisely. Show all work - credit will not be given for numerical solutions that appear without explanation. **[Total 63=3+60 pts]**

Grading Rubric

Each question is worth 3 points and will be graded as follows:

3 points: Correct answer with work shown

2 points: Incorrect answer but attempt shows some understanding (work shown)

1 point: Incorrect answer but an attempt was made (work shown)

0 points: Left blank or made little to no effort/work not shown

Reflective Journal [3 pts]

(link to your live google doc)

<https://docs.google.com/document/d/1dzAggy7UhyrmR9CylVDsXdToMj1JAtrVwGZaqPQtogo/edit?usp=sharing>

I. Collecting Data: Sampling (9 x 3 = 27 pts)

Your local movie theater has put in a lot of work to make their customer's movie going experience great. They would like to take a random sample of their recent customers and email out a survey asking about their experience.

1) For the following situations, identify the bias and explain how they would introduce bias into the results.

a) You send out an email to the first 10 customers that buy tickets to a movie.

Answer:

This is an example of undercoverage bias introduced by a convenience sample. The first people to buy tickets to a movie are more likely to be avid movie-goers compared to the average person. This would exclude everyone who buys a ticket last minute.

b) Of the 10 movies that are playing in a single day, you randomly select one movie and email the survey to everyone who attended that movie.

Answer:

This is another example of undercoverage bias introduced by a convenience sample. By randomly selecting a movie, you are limiting the respondents to fans of a certain genre, and potentially excluding certain people based on the time of day they go to the movies depending on the show times of the movie.

c) You set up a sign as the patrons exit the theater and tell patrons to email support@movies.com if they would like to share their movie going experience.

Answer:

This is an example of voluntary response bias. You will only get respondents who are highly invested in the movie-going experience or with very strong opinions. This would lead to an underrepresentation of people who don't have strong feelings or opinions to share.

d) You have the manager of the movie theater waiting outside the movie theater and interviewing every 10th person who walks out about their movie going experience.

Answer:

This is an example of an observer bias. The patrons are more likely to give positive responses to an employee of the movie theater who might have influence over their future visits to the movie theater if they give a bad review.

You obtain a list of the following data in a month at the movie theater: Order ID, Date, Customer Name, Customer Email, Movie Title, Show Time, Number of Tickets Purchased.

2) You would like to email out a survey to a sample of customers. For the following sampling methods, describe how you would use the information above to select a sample.

a) Simple Random Sample

Answer:

I would use a random number generation method to randomly select data based on Order ID.

b) Stratified Random Sample

Answer:

I would use the Movie Title to create strata, which should provide the best representation of different kinds of movie-goers.

c) Cluster Random Sample

Answer:

I would use the Customer Name to create clusters. Customers with the same first name could be clustered, then random clusters are selected for sampling. This would introduce the least bias compared to clustering on other variables.

d) Systematic Random Sample

Answer:

I would order the data by Customer ID, then select every 10th customer. This would introduce the least bias compared to the other attributes.

e) A Census

Answer:

I would email everyone in the data set.

II. Collecting Data: Selecting Random Samples (11 x 3 = 33 pts)

1) The local university is interested in how many of its students plan on using some of the new facilities it installed over the summer. There are 3000 students who attend the university and they want to send out a questionnaire to 100 of them. They decide to take an SRS from an alphabetized list of students.

(a) Describe how you will select the sample.

Answer:

I would number the students from 1 to 3000, then use a table of random digits to select 100 random numbers in that range. I will do this by looking at every 4 digits and determining if it falls within the range and is not already selected. If so, I will add it to the list of random numbers; if not, I will move on to the next 4 digits. Once I have 100 numbers, I will use them to select corresponding students from the list and send them the questionnaire.

(b) Below is a sample of the table of random digits. Using the table and then method described above, what are the first four numbers that will be part of your sample?

23328 99330 01231 42492 73831 02911 01524 32932 34334 74280 29357 29301

Answer:

2332

2314

2492

1029

2) Your school has a pep rally in the gymnasium. The following is a map of the gym, where each box represents a seat that a student sits in.

1	2	3	4	5	G y m F i o r	301	302	303	304	305
...	...	...	...	...		...	...	...	...	...
146	147	148	149	150		446	447	448	449	450
151	153	154	155	156		451	452	453	454	455
...	...	...	...	...		...	...	...	...	...
296	297	298	299	300		596	597	598	599	600

Freshman: Seats 1 – 150

Sophomores: Seats 151 – 300

Juniors: Seats 301 – 450

Seniors: Seats 451 – 600

Describe how you would use a random number generator to select 40 students to complete a survey with each of the following sampling methods. **NOTE: I want you to describe it, not actually do it!*

(a) Simple Random Sample

Answer:

I would use the random number generator to select 40 random numbers between 1 and 600 using the method previously described. I would then select students sitting in those seats to complete the survey.

(b) Stratified Random Sample

Answer:

I would use the random number to create four sets of ten random numbers between 1 and 150. It's okay if there is overlap between the sets of random numbers. I will then add 0 to every number in the first set, add 150 to every number in the second set, add 300 to every number in the third set, and add 450 to every number in the fourth set. I will then combine the four sets of random numbers and select all of the students sitting in those seats to complete the survey.

(c) Cluster Sample

Answer:

I would use the random number generator to select 8 random numbers between 1 and 120, then multiply those random numbers by 5. I would then add to the list of numbers all of the numbers 1 below, 2 below, 3 below, and 4 below the numbers already in the list. I would then select the students sitting in those seat numbers to complete the survey.

3) [A study was done](#) to determine if eating chocolate during pregnancy has an effect on infant temperament at the age of 6 months. The study asked about 300 pregnant women to report their chocolate consumption (Never/Seldom, Weekly, and Daily). Six months after they had given birth, the researchers then asked the mothers to report their child's overall temperament (activity, smiling, fear, etc.). They determined "Maternal prenatal consumption of chocolate was associated with the infant temperament in this study. Mothers who reported consuming chocolate daily rated their infants at 6 months as more positively reactive and active". Was this an observational study or an experiment? Explain.

Answer:

This is an observational study because the researchers did not attempt to influence the responses in any way. The mothers were free to eat as much chocolate as they wanted and were able to switch between groups of chocolate consumption during the study.

4) [A large study](#) in the UK was done to determine if there was a link between cell phone usage and brain cancer. The study set out to determine if using a cell phone more, increased the likelihood of getting brain cancer. The study took about 800,000 women and asked them to report their cell phone use in 1999, in 2005, and again in 2009. In 2009, they recorded the proportion of women who had developed brain cancer. Their conclusion was "In this large prospective study, mobile phone use was not associated with increased incidence of glioma, meningioma or non-CNS cancers." Is this an observational study or an experiment? Justify your answer.

Answer:

This is an observational study because the researchers did not attempt to influence the responses in any way. In fact, because it was a prospective study, it was impossible for the researchers to alter the treatment level.

5) McGraw Hill would like to test synchronous vs asynchronous learning for a new Algebra I classroom. They give a test to a group of 200 students who are brand new to the Algebra I classroom. They then assign half of the students to study Algebra I synchronously (with a classroom and traditional teacher) and half of the students to study Algebra I asynchronously (online and at their own pace). At the end of the year, students were given the same test and the difference between the first test and the second test were compared. Is this an observational study or an experiment? Justify your answer.

Answer:

This is an experiment because the students were split into groups by the researchers. This is an attempt by the researchers to influence the response by assigning study participants to groups with different treatments. In this case the treatment is synchronous vs asynchronous learning.

THE END